



MultiHop Configuration Software Manual

Original Instructions

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Chapter 1 MultiHop Configuration Software

Use Banner's MultiHop Configuration Software to view your MultiHop radio network and configure the radio and its I/O.

MultiHop Configuration Software Network and Device Overview screen

Network

Configuration

Reprogram

Register View

Settings

Network and Device Overview

Network Query

Master address: 1

Device address

Read

Site Survey

Devices: 24

Repeaters: 1

Slaves: 22

Unreachable: 2

Save to File

Name	Role	Modbus Address	Device Address	Parent Address	Signal Strength	Green	Yellow	Red	Misses	Serial Number	Model Number	Build Date	RF FW P/N	RF EE P/N	RF FW Ver	RF EE Ver	LCD FW P/N	LCD EE P/N	LCD FW Ver	LCD EE Ver
Master 900MHz HES	Master	1	23846	23846	0	0	0	0	0	154918	186215	001544	175068	3.6C	175070	1.0				
DATA RADIO DEVICE	Slave	35	34520	23846	50	0	50	0	50	100056	000000	000000	165062	3.0E	159481	0.2A				
DATA RADIO DEVICE	Slave	17	24200	23846	0	0	0	0	0	155272	151687	001544	168893	3.4	157721	1.1				
Multihop Data Radio	Slave	14	64179	23846	0	0	0	0	0	195251	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	45	63129	23846	0	0	0	0	0	259737	151687	001415	168893	2.6	157721	1.1				
DATA RADIO DEVICE	Slave	19	24203	23846	0	0	0	0	0	155275	151687	001544	168893	3.4	157721	1.1				
DATA RADIO DEVICE	Slave	90	4775	23846	0	0	0	0	0	135847	183420	001523	168893	2.6	157721	1.1				
Multihop Data Radio	Slave	15	64180	23846	0	0	0	0	0	195252	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	37	56005	23846	0	0	0	0	0	842437	190055	1541	169345	3.1	169449	0.1C				
Multihop Data Radio	Slave	16	64154	23846	0	0	0	0	0	195256	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	20	24196	23846	0	0	0	0	0	155268	151687	001544	168893	3.4	157721	1.1				
DATA RADIO DEVICE	Slave	36	56006	23846	0	0	0	0	0	842438	190055	1541	169345	3.1	169449	0.1C				
MH MGate SID 13	Slave	13	64176	23846	0	0	0	0	0	195248	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	18	24202	23846	0	0	0	0	0	155274	151687	001544	168893	3.4	157721	1.1				
DATA RADIO DEVICE	Slave	27	9619	23846	0	0	0	0	0	271963	151687	001425	168893	2.6	157721	1.1				
Multihop Radio H12	Repeater	91	58281	23846	78	70	0	0	22	123817	151685	1512	148691	2.2	151698	1.3	136499	3.2	148880	1.0
DATA RADIO DEVICE	Slave	84	4794	58281	0	0	0	0	0	135866	183420	001523	168893	2.6	157721	1.1				
DATA RADIO DEVICE	Slave	32	9621	58281	0	0	0	0	0	271965	151687	001425	168893	2.6	157721	1.1				
MH MGate SID 12	Slave	12	64185	58281	0	0	0	0	0	195257	157598	001233	157719	2.2	157722	1.0				
Multihop Data Radio	Slave	78	29005	58281	0	0	0	0	0											
DATA RADIO DEVICE	Slave	31	65198	58281	0	0	0	0	0	261806	151687	001417	168893	2.6	157721	1.1				
DATA RADIO DEVICE	Slave	82	4744	58281	0	0	0	0	0	135816	183420	001523	168893	2.6	157721	1.1				
MH MGate SID 11	Slave	11	64181	58281	0	0	0	0	0	195253	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	83	4743	58281	0	0	0	0	0	135815	183420	001523	168893	2.6	157721	1.1				
Unreachable devices addresses																				
		Device Address	Reprocess																	
		4776																		
		29001																		

The software connects to a MultiHop client radio using one of four methods.

- Serial; using a USB to RS-485 (for RS-485 radios) or a USB to RS-232 (for RS-232 radios) converter cable.
- Modbus TCP; using an Ethernet connection to an Ethernet radio client.
- Serial DXM; using a USB cable to a DXM Controller to access a MultiHop client radio.
- TCP DXM; using an Ethernet connection to a DXM Controller to access a MultiHop client radio.

Banner recommends using **BWA-UCT-900**, an RS-485 to USB adapter cable with a wall plug that can power your radio while you configure it. The adapter cable is not required when connecting to a DXM Controller.

Download the most recent software revision from the Wireless Reference Library on Banner Engineering's website: www.bannerengineering.com.

Installation

Use the MultiHop Configuration Software to configure the radios within your MultiHop radio network. The MultiHop Configuration Software uses a USB to RS-485 converter cable to connect a MultiHop radio to a USB connection on a computer.

Installing the Software—Before connecting the USB to RS-485 adapter cable, install the MultiHop Configuration software downloaded from the Banner Engineering website. To ensure a complete installation, verify you have an active Internet connection or .NET 4.5 (or higher) before installing the software.

BWA-UCT-900 Adapter Cable



Connecting the MultiHop Radio using the USB/Converter:

1. Complete the software installation.
2. Connect the USB end of the Banner USB converter to a USB port on the computer. The computer should recognize the Banner USB converter device.

3. Connect a Banner MultiHop radio to the M12 connector end of the USB converter. The MultiHop radio powers up. When using a third-party USB converter, manually wire the communication and power connections. Refer to Banner's MultiHop radio datasheet for the wiring pin-out configurations.
4. Launch the MultiHop Configuration Software.
5. Select the USB from the pull-down list.
6. Use the **Device > Connection Settings** menu to select a port. The factory default baud rate is 19.2k, 1 stop bit, no parity.

Connecting to a MultiHop Radio using a DXM Controller:

1. Connect a USB cable from a USB port on the computer to the USB port on the DXM Controller.
2. For Ethernet, connect an Ethernet cable from the DXM Controller to an Ethernet switch.
3. Cycle power on the DXM Controller after the Ethernet cable is attached to the DXM Controller.

For most of the MultiHop Configuration Software screens, a grayed box represents a parameter read from the device, the control is inactive, or the software is reading a field. A white box represents parameters entered by the user.

Menu Bar

The **File** and **Device** menus select the basic file operations and communication configuration for the DX80 Gateway.

Access the **File** and **Device** menus from the menu bar.

File Menu

Exit—Closes the COMM port and exits the program. Any data that is not saved is lost.

Load—Loads a specific XML configuration file into the tool for editing. This XML configuration file is unique to the MultiHop Configuration Software and is not compatible with other configuration software XML files.

New—Loads blank configuration settings into the tool.

Save—Saves the current settings in the configuration software under the current XML file name. The program switches to Save As if there is no XML file name assigned.

Save As—Saves the current settings in the configuration software to a new file name specified by the user.

Device Menu

Traffic Watcher—Views all serial data traffic between the host and the connected radio. The Traffic Watcher logs file loads, error messages, and data transfers.

Connection Settings—Sets the COMM port to communicate with the connected radio. There are four possible connections with the MultiHop Configuration Software.

Serial—Direct connection to the client radio using a RS485 to USB converter cable.

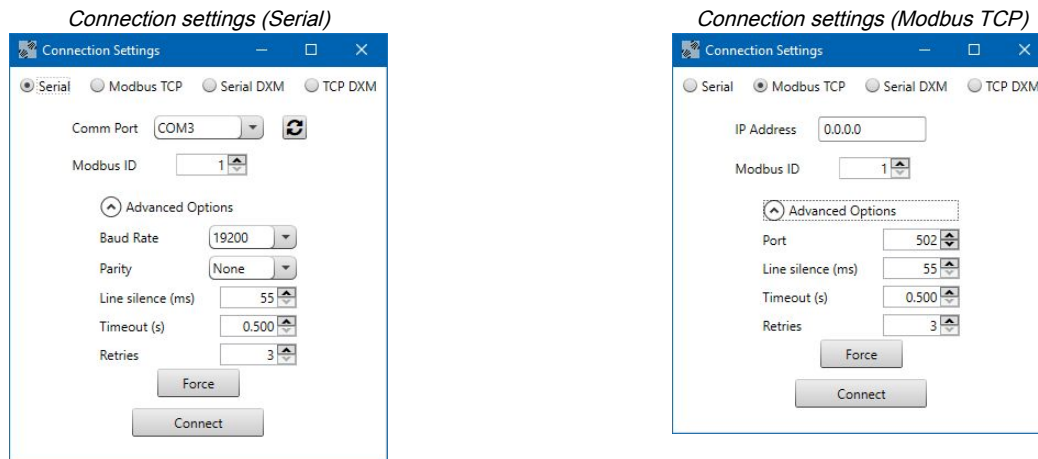
Serial DXM—Connecting to a DXM internal ISM radio using USB

TCP DXM—Connecting to a DXM internal ISM radio using Ethernet; select the VPN checkbox only when your DXM is connected to the Internet via a VPN

Connection Settings

The configuration software connects directly to a standalone parent radio using the USB to RS-485 adapter cable. When using a DXM Controller, connect to the internal radio through the DXM Controller.

After the software launches, go to **Device > Connection Settings** to choose one of the four communication modes: **Serial**, **Modbus over TCP**, **Serial DXM**, or **TCP DXM**.



Basic Connection Settings

Comm Port / IP Address—Select the COMM port of the PC or, when using Ethernet, the IP address of the device. Click the refresh button to refresh the COMM ports on the PC.

Connect/Click to Cancel—Click to verify a radio is present by reading back the RF firmware. After a connection is in progress, this button changes to **Click to Cancel**.

Modbus ID—Modbus address ID of the radio.

Modbus TCP—Use when connecting to the parent radio using an Ethernet connection. The IP address must be known to communicate with the device.

Serial—Use when directly connecting to the parent radio using the USB to RS-485 adapter cable.

Serial DXM—Use when configuring an internal radio of the DXM Controller using the USB connection on the DXM Controller.

TCP DXM—Use when configuring the internal radio of the DXM Controller using the Ethernet port. The standard DXM TCP port is 8844.

Advanced Connection Settings

Baud Rate (Serial only)—Defines the baud rate for the serial connection of the USB to RS-485 converter or DXM USB connection. Default of 19200 for the USB to RS-485 converter or 115k for the DXM USB connection.

Force—Click to have the software use the selected parameters and force a connection to a device without validating the device is present.

Line Silence—Specifies the wait time between two consecutive Modbus messages created by the DXM Controller. Too short of a wait time may cause external Modbus devices to miss or create errors on Modbus traffic. A typical setting should be 50 to 100 ms.

Master Ping—For most applications, leave the Master Ping set to 0. To communicate directly with a DXM baseboard, set the value to 2. Using Master Ping allows you to have multiple client radios on a single network. For more information about this advanced setting, contact Banner Engineering Corp and ask to speak to a Wireless applications engineer.

Parity (Serial only)—The default parity setting is None. Optional settings are Even or Odd.

Port (Ethernet only)—Defines the Ethernet port used when communicating via Modbus/TCP or to the DXM Controller. The default is 502.

Retries—The number of times the software resends Modbus messages before it errors out. The minimum number of retries is three.

Timeout—The time allowed for a Modbus message to complete. The default [timeout interval](#) is 0.5 seconds for a serial connection and 5 seconds for a TCP connection. This may need to be extended for battery-powered devices or networks with communications paths that include multiple repeater devices.

Chapter 2 Software Screens

The following sections explain the function of each screen.

Network and Device Overview Screen

The **Network and Device Overview** screen displays an organized view of the MultiHop radio network, noting which devices are the client, repeaters, and servers.

The network structure is indicated by the format of the list. MultiHop radios are indented to indicate which parent they are communicating with. In addition to displaying the structure of the data radio network, signal quality information is displayed in a color-coded format for easy viewing.

Network

Configuration

Reprogram

Register View

Settings

Network and Device Overview

Network Query

Master address: 1

Device address

Read

Site Survey

Devices: 24 Repeater: 1 Slaves: 22 Unreachable: 2

Save to File

Name	Role	Modbus Address	Device Address	Parent Address	Signal Strength	Green	Yellow	Red	Misses	Serial Number	Model Number	Build Date	RF FW Ver	RF EE Ver	RF EE PN	LCD FW Ver	LCD EE Ver	LCD EE PN	LCD EE Ver	
Master 900MHz HES	Master	1	23046	23046	0	0	0	0	0	154918	190215	001544	175069	3.0C	175070	1.0				
	DATA RADIO DEVICE	Slave	35	34520	23046	50	0	0	50	100056	000000	000000	165062	3.0E	159481	0.2A				
	DATA RADIO DEVICE	Slave	17	24200	23046	0	0	0	0	155272	151687	001544	169893	3.4	157721	1.1				
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	DATA RADIO DEVICE	Slave	84	4794	58281	0	0	0	0	135866	183420	001523	169893	2.6	157721	1.1				
	DATA RADIO DEVICE	Slave	32	9821	58281	0	0	0	0	271965	151687	001425	169893	2.6	157721	1.1				
	MH MGate SID 12	Slave	12	64185	58281	0	0	0	0	195257	157568	001233	157719	2.2	157722	1.0				
	MultiHop Data Radio	Slave	78	29005	58281	0	0	0	0					169893	2.6	157722	1.1			
Unreachable devices addresses	DATA RADIO DEVICE	Slave	31	65198	58281	0	0	0	0	261806	151687	001417	169893	2.6	157721	1.1				
	DATA RADIO DEVICE	Slave	82	4744	58281	0	0	0	0	135816	183420	001523	169893	2.6	157721	1.1				
	MH MGate SID 11	Slave	11	64181	58281	0	0	0	0	195253	157568	001233	157719	2.2	157722	1.0				
	DATA RADIO DEVICE	Slave	83	4743	58281	0	0	0	0	135815	183420	001523	169893	2.6	157721	1.1				

Unreachable devices addresses

Device Address

4776

29001

Reprocess

Device Address—The device address (DADR) is a unique number based on the device's serial number and is set by the factory. This address displays as part of the RUN menu's auto-loop display on the device's LCD screen while the radio is operating.

Device Role—The devices are listed as client (only one per network), Repeater, or server. Set the device type using each radio's DIP switches.

Client—The client radio controls the overall timing of the network and is always a parent device for other MultiHop radios. The host system connects to this client radio.

Repeater—When a MultiHop radio is set to repeater mode, it acts as both a parent and a child. The repeater receives data packets from its parent, then re-transmits the data packet to the children within the repeater's network.

Server—The server radio is the end device of the data radio network. A MultiHop radio in server mode does not re-transmit the data packet on the radio link.

Modbus Address—Set the MultiHop **device ID** using the radio's rotary dials. Note that the MultiHop ID is also the Modbus ID. The Modbus Address (Modbus ID) must be a unique number to access I/O point data on the radio device. For radio devices that do not have I/O and do not want to be individually accessed, a Modbus ID is not required.

Model Number—The model number is a 6-digit number used by the manufacturer to identify the device model of the radio.

Name—Use the name string to identify the radio device. The name is saved in the MultiHop device. Change the name using the **Configuration** screen.

Parent Address—The client radio and repeater radios act as a parent device to server radios. The parent radio provides network timing and data routing to server radios.

Radio (RF) and LCD Firmware (FW) and EEPROM (EE)—These parameters are set at the factory. You may be asked for this information if you call Banner Engineering for technical support. To update these programs, obtain the newest versions from the factory, then use the **Reprogram** screen to copy these files to your radio.

RF FW—Part number and version number for installed firmware of the radio microprocessor.

RF EE—Part number and version number for installed EEPROM of the radio microprocessor.

LCD FW—Part number and version number for installed firmware of the LCD microprocessor.

LCD EE—Part number and version number for installed EEPROM of the LCD microprocessor.

Save to File—Saves the information displayed on the screen (device name, type, address, etc) to a text file.

Serial Number—The serial number is a 6-digit, unique number that identifies that specific radio. No other radio will be assigned this number. Each Sure Cross radio uses its serial number to generate the device address that displays on the **Network and Device Overview** screen and the radio's LCD.

Signal Strength—The number of packets successfully transmitted, during a [site survey](#), at the various signal strengths are represented by the "green," "yellow," and "red" delineations.

Green—a strong radio signal

Yellow—a good radio signal

Red—a marginal radio signal

Misses—represents the number of packets not received on their first transmission

Unreachable—The unreachable devices are radios the client radio has detected but was unable to locate during the last scan. From the Unreachable drop-down list, select a device or range of devices and click **Reprocess**. The client radio will attempt to connect to and read this device.

Configuration Screens

The **Configuration** main tab allows you to configure your device by defining device-level parameters and I/O parameters. Use the **User Defined Devices** screen to create a user-defined device for use in your radio network.

Configure Device Screen

Use the **Configure Device** screen to define the default conditions, push registers, [flash patterns](#), Modbus parameters, device restore conditions, and to remap registers.

Enter the [device ID](#) of the MultiHop radio you want to configure. To select a MultiHop radio by its unique device address (Not Modbus ID), select the Device Address checkbox and then enter the Device address.

Manual Mode—Select the **Manual Mode** checkbox and select the **Device Type** from the drop-down list. Selectable device types are:

H—M-H (no I/O) models
 H1—M-H1 models (including SPRFs)
 H2—M-H2 models (including SPRFs)
 H3—M-H3 models (including SPRFs)
 H4—M-H4 models (including SPRFs)
 H5—M-H5 models (including SPRFs)
 H6—M-H6 models (including SPRFs)
 H12—M-H12 models (including SPRFs)
 H14—M-H14 models (including SPRFs)

H1B—M-H1B Board Modules (including SPRFs)
 DXM100x1—All DXM100x1 models
 H2B—M-H2B Board Modules (including SPRFs)
 DXM100x2—All DXM100x2 models
 DC-LATCH—MultiHop DC-LATCH models
 DXM150x1—All DXM150x1 models
 HM—MultiHop M-GAGE models
 DXM150x2—All DXM150x2 models
 U—MultiHop Ultrasonic and Ultrasonic/Light models

To read the existing parameters, click **GET All Parameters**. To write changes to the selected device, click **SEND All Parameters**.

Reading Configuration Data from the MultiHop Radios

To read, or "get" configuration data from a MultiHop radio anywhere in the network, enter the MultiHop Modbus ID and click the appropriate **GET** button.

NOTE: The MultiHop Modbus ID is assigned to each device using the rotary dials on the front of all MultiHop radios.

Name

Assign a name to each device to it easier to track the device in the configuration software.

The **Name** must be less than 18 characters and contain only the standard alpha-numeric characters, the following standard ASCII characters: *, +, -, /, <, >, or a space.

After making any changes, click **SEND all parameters and IO** to write all the changes. The status bar at the bottom of the screen indicates the status of the process. Green (Communication Status Ready) indicates the process is complete. Red (error) indicates an error. Yellow (Getting Data) indicates the network is retrieving the requested data.

Default Conditions

Default conditions are the conditions under which outputs are sent to their defined [default output value](#). These conditions apply to all outputs.



When a radio is **Out of Sync** (experiences an [out of sync radio link loss](#)), it is not communicating with its parent radio. Selecting the Out of Sync condition sets all outputs (on this device) to their specific default values when this radio has lost its communication link with its parent radio. The [default output values](#) are selected under each of the output parameter sections of this screen.

Selecting **Start Up** sets this radio's outputs to their default values when this radio is powered up.

A **Comms Timeout** refers to the communication between the host system and the selected radio (which may be any radio within the network). Selecting Communication Timeout sets this radio's outputs to their selected default values when the host system has not communicated with this radio within the time specified. Set the communication timeout in seconds.

Selecting **Out of sync power save** sets the radio to low power mode if the radio is out of sync for more than 5 minutes. The radio wakes up every 5 minutes to search for a parent radio. This setting is recommended for battery-powered devices.

After making any changes, click **SEND all parameters and IO** to write all the changes. The status bar at the bottom of the screen indicates the status of the process. Green (Communication Status Ready) indicates the process is complete. Red (error) indicates an error. Yellow (Getting Data) indicates the network is retrieving the requested data.

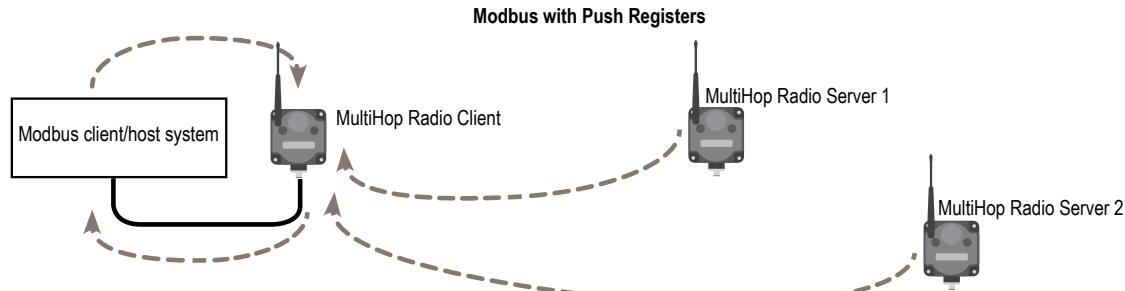
After the radio problems are resolved with the parent radio or communication with the host system has resumed, the host system is responsible for writing the operating output value.

Push Registers

Push registers automatically transmit register data from a MultiHop device back to the client MultiHop radio.

In the MultiHop system, every device is treated as a separate unit. In the Modbus protocol, every device is a Modbus server device. To get I/O register information, a host system must interrogate each Modbus server one at a time. This is the standard mode of operation of MultiHop devices and works well for most applications.

Push registers are a feature of the MultiHop system. The user can define up to four registers per MultiHop I/O device that transmit data to be stored at the client MultiHop data radio. This allows a host system to interrogate only the client device for I/O information without dealing with the latency of the wireless link.



Using the push registers is beneficial in applications where:

- End devices are battery powered with infrequent changes, but notification of changes wants to be immediate. Instead of a host continually polling a device for I/O changes, which consumes battery power, push registers transmit data back to the client device when it changes and/or on a periodic cycle.
- A host polling loop becomes impractical because of the time required to communicate with each remote device individually (large networks). The data changes are usually infrequent and the devices are typically battery powered. With large networks, push registers allow each end device to transmit data to the client when it changes and does not rely on the host request to initiate a transaction.

Using push registers is not beneficial in typical monitoring applications with periodic data requirements of a minute or longer.

A side benefit of using the push registers is the ability to use a device status register for each MultiHop device. With a push register interval defined, a **client Sweep Interval** can be defined to verify the communication of each device pushing data back to the client device. During the client sweep interval, every device must communicate with the client device or the status register will be zero (missed communications).

06800s Push Registers

The following are the push register parameters for each radio. Push register values are stored on the radio client at registers 7909 through 7912, by MultiHop radio ID.

6831 Input Push Register 1

6832 Input Push Register 2, etc through 6850

For a server or repeater, these define which registers to push to the client device. This allows a server/repeater to send local input data back to the client without having to wait to be asked for the data.

The Modbus register value should be Modbus Register minus 40001. For example, to push register 40001, write a value of 40001 minus 40001, or 0. To push register 40002, write a value of 40002 minus 40001, or 1.

6871-6872 Push Register Report Interval

Push registers can report on a periodic interval or on [change of state](#). The [report interval or report rate](#) establishes how often, in frames/slots, to push data to the client. 6871 is the high word and 6872 is the low word. This is typically a server or repeater parameter.

Set a periodic interval using 40 millisecond units. For example, to set the report interval to 1 minute (60 seconds ÷ 0.040 seconds = 1500 units), write a 0 to register 46871 = 0 and write a 1500 to register 46872.

Enable change of state reporting by writing a 1 to register 43004 for input 1 or register 43024 for input 2. Other inputs have different configuration registers to enable change of state reporting.

6873-6874 Client Sweep (Health Heartbeat) Interval

Works in conjunction with each server/repeater push register report interval (6871–6872) to create device status information in the client device. This allows the user to monitor the health of each device in the MultiHop network by defining how often the client should expect each repeater and server to push register values back to the cached copies in local storage on the radio client.

This health check should be at least 10 frames longer than the push register report interval. Use the registers to set a periodic interval using 40 millisecond units.

For example, to set the client sweep interval to 1 minute (60 seconds ÷ 0.040 seconds = 1500 units), write a 0 to register 6873 = 0 and write a 1500 to register 6874; register 6873 is the high word and register 6874 is the low word.

Read the individual status registers for each Modbus server device by sending a Modbus read command to the particular server device at register 7904. For example, to read the status register of server 24, send a Modbus read command to server ID 24 register address 7904, the contents will be either 0 (bad) or 1 (good). The individual status data can be read by device address (not Modbus address) starting at Modbus register 12701. The client radio register 12701 holds the number of devices in the system. The following registers 12702+ contain the status information for those devices.

A bit-packed version of the status data is available starting at register 12981. The client radio register 12981 holds the number of device in the system, register 12982 holds the status bit fields for devices 0-15, 12983 holds the status bit fields for devices 16-31.

When dealing with device addresses, device order is based on the order they first sync to the client device. That ordered list is stored starting at Modbus registers 7001 in the client radio. Modbus register 7001 holds the count of the number of devices in the network, with the following registers 7002 and greater holding the device address of the MultiHop radio. The order of this list defines how the status bits or words are stored in the client radio.

6875 Report Interval Random Modulus

The Report Interval Modulus parameter provides an offset to the report interval between 0 and the [report interval or report rate](#) modulus to minimize the collisions that occur if multiple radios are reporting back at the same time to the radio client.

Enter this parameter as the number of 40 millisecond frames for the offset. A value of 5 represents 200 milliseconds, or 5×0.040 seconds.

For example, if the reporting interval is set to every minute and the report interval modulus is set to 3, the report interval will be every minute to a maximum of every minute + 120ms ($3 \times 40\text{ms}$).

IO Pulse Patterns

Setting the I/O pulse pattern establishes an on and off pattern that can be used for a discrete output or [switch power](#).

Define **I/O Pulse Patterns** by selecting specific timeslots to turn the output on or off. Although the pulse pattern was originally designed to turn on and off an indicator light, the pulse pattern can be set for any discrete output or switch power. Users may configure up to four different patterns per radio device. In the example shown, Pattern 1 is configured to turn on for the first 240 milliseconds within a 1280 millisecond range.

After making any changes, click **SEND all parameters and IO** to write all the changes. The status bar at the bottom of the screen indicates the status of the process. Green (Communication Status Ready) indicates the process is complete. Red (error) indicates an error. Yellow (Getting Data) indicates the network is retrieving the requested data.

Modbus Parameters

Enter the first Modbus address and the total number of MultiHop devices in the network. These parameters only apply to the client MultiHop radio.

Device Restore

Use these commands to restore default settings to the selected device.

To restore default device parameters:

1. At the top of the screen, select a MultiHop device.
2. Click on **GET All Parameters** to get that device's I/O configuration.
3. In the **Device Restore** section, select the appropriate checkboxes.
 - **Restore Default I/O Configuration.** Restores default I/O configuration parameters to the values indicated in the device's datasheet, such as sample and report rates.
 - **Restore Default System Parameters.** Restore default system parameters, such as the radio configuration and binding code.

- **Re-Initialize Variables from Serial Number.** Restores the default values of all variables calculated from the serial number. This is not typically recommended.
4. Click on **Restore Device** to restore the selected default values to your device.

Remap Registers or Register Aliasing

Use the **Remap Registers** section of the **Configure Device** screen to map registers to contiguous register locations to optimize Modbus read/write functions.

Alias Registers (601-616)		Register Content (101-116)	
1	5	101	0
2	7	102	0
3	9	103	0
4	10	104	0
5	8	105	0
6	501	106	0
7	502	107	0
8	1	108	0
9	1	109	0
10	1	110	0
11	1	111	0
12	1	112	0
13	1	113	0
14	1	114	0
15	1	115	0
16	1	116	0

1. Verify the desired MultiHop Radio **Address** is selected in **Configuration Address**.
2. Click the arrow next to **Remap Registers**.
3. Fill in the source registers you would like to alias. The **Alias Registers** rows 1 through 16 are the user-defined entries of addresses of the registers to alias (rearrange). This alias table is stored in the MultiHop radio register addresses 601 through 616. In the example, source registers addresses 5, 7, 9, 10, 8, 501, and 502 are entered into the table.

The aliased **Register Contents** are in registers 101 through 116. For this example, when a host system reads Modbus registers 101 through 107 of the MultiHop radio, the register contents come from register 5, 7, 9, 10, 8, 501 and 502.

Get I/O Points

To configure the inputs and outputs for each MultiHop radio, enter the radio ID at the top of the screen and click **GET all parameters and IO**. All inputs and outputs available for the select device display on the screen.

Inputs		Outputs	
Discrete Input 1	Enabled	Discrete Out 1	Enabled
Discrete Input 2	Enabled	Discrete Out 2	Enabled
Analog Input 1	Enabled	Switch Power Output 1	Enabled
Analog Input 3	Enabled	Switch Power Output 2	Enabled
Thermister	Enabled		
Counter Input 1	Enabled		
SDI-12 Input 1	Enabled		
SDI-12 Input 2	Enabled		
SDI-12 Input 3	Enabled		
SDI-12 Input 4	Enabled		
SDI-12 Input 5	Enabled		

The screenshot displays two side-by-side configuration windows. The 'Inputs' window on the left shows 'Discrete Input 1' as 'Enabled' with a sample rate of '00:00:00.04'. It includes a 'Digital Configuration' section with 'Input type' set to 'NPN', 'Sample high' and 'Sample low' both at '0', and checkboxes for 'Push on change of state' (checked) and 'Enable active time counter' (unchecked). Below is a 'Switch Power Settings' section with checkboxes for 'SP1' and 'SP2' (both unchecked), a 'Warm up' time of '00:00:00.00', and a 'Voltage' of '0'. There is also an 'Extended Input Read' section with a checked 'Continue sampling when out of sync' option. 'Discrete Input 2' is shown as 'Disabled'. The 'Outputs' window on the right shows 'Discrete Out 1' as 'Enabled'. It has an 'Output Configuration' section with 'Use flash pattern' set to '1'. Below is a 'Discrete Configuration' section with checkboxes for 'Hold last state' (unchecked), 'Default output value' set to '0', and 'Invert output' (unchecked). At the bottom, 'Discrete Out 2', 'Switch Power Output 1', and 'Switch Power Output 2' are all shown as 'Disabled'.

Select the device you want to configure by using the drop-down list to select the device address. Click **GET all parameters and IO** to retrieve any existing configuration data from the device.

The following commands are general to the entire screen.

Address

Select the address of the device you want to configure. Click **GET** to retrieve that device's information before making changes to it.

Enabled

Turns on any specific I/O point. When unselected, the specific I/O point is disabled for the MultiHop radio device selected at the top of the screen

GET

Click **GET all parameters and IO** to read all system and IO parameters from the radio. Click **GET system parameters** to read only the system parameters. Click **GET all IO points** to read only the IO point parameters. Click **GET** to read the parameters for that specific IO point.

Name

Enter a name for the device. This name displays on the other configuration screens.

After making any changes, click one of the following to write the changes. The status bar at the bottom of the screen indicates the status of the process. Green (Communication Status Ready) indicates the process is complete. Red (error) indicates an error. Yellow (Getting Data) indicates the network is retrieving the requested data.

- **SEND all parameters and IO**—Writes all changes to all parameters and IO to the network
- **SEND system parameters**—Writes only the changes to the system parameters to the network
- **SEND all IO points**—Writes only the changes to the IO point parameters
- **SEND**—Writes the parameters only for that specific IO point

Inputs

The following parameters are used to configure the inputs.

The input parameters vary, depending on the enabled and available input types.

Input Configuration

Select **Enable** to turn on this input.

Sample Rate

The [sample interval or sample rate](#) defines how often the Sure Cross device samples the input. For battery-powered applications, setting a slower rate extends the battery life. Set the sample rate/interval in hours:minutes:seconds.

Analog Configuration

Enable push on change of state

Enables push registers for this input. When the discrete input changes state, the register value is pushed to the client radio if this register is configured to be a push register. For analog inputs, use the threshold and hysteresis parameters to define "on" and "off" points.

Enable register full scale

Turning Fullscale ON sets the entire register range of 0 through 65535 to represent the selected minimum through maximum input values. With Fullscale turned on, a register value of 0 represents the selected minimum value in microamps (for current inputs). A register value of 65535 represents the selected maximum value in microamps. For example, a register value of 0 is 0 and the register value of 65535 represents 20 mA (or 20,000 microamps). With Fullscale turned OFF, the register value represents unit-specific input readings. For units of current (mA), register values are stored as microAmps. Voltage values are stored as millivolts. A sensor reading of 15.53 mA is stored as 15530.

Counter Options

Counter type

Select the counter type: event or frequency. The **frequency counter** calculates the frequency of the input signal, in Hz. The **event counter** counts the total number of times an input

signal changes to the high/ON/1 state. The [frequency counter](#) is typically used to measure flow rates, such as measuring the flow rate of items on a conveyor or the speed at which a windmill spins. The [event counter](#) can be used to measure the

total operational cycles of a spinning shaft or the total number of items traveling down a conveyor.

Digital Configuration

Configure the following parameters for discrete/digital inputs.

Continue sampling when out of sync (discrete)

Select to have the MultiHop device continue sampling the I/O even when the radio is out of sync with its parent radio. This is particularly valuable on counter inputs.

Enable active time counter (Discrete)

Select the checkbox to have the device count the length of time the discrete input is held high.

Input type (Discrete)

Select either NPN or PNP discrete input types.

Preset Value

Enter a number in the selection box and press the Set Value button to write a preset counter value to the register.

Sample High or Sample Low

For discrete inputs, the sample high parameter defines the number of consecutive samples the input signal must be high before a signal is considered active. Sample low defines the number of consecutive samples the input signal must be low before a signal is considered low. The sample high and sample low parameters are used to create a filter to avoid unwanted input transitions. The default value is 0, which disables this feature. The value range is 1 through 255.

Sample Out of Sync

When a device is not in sync with the wireless network, it will not sample its I/O points. If **Continue Sampling When Out of Sync** is selected, the device continues to sample the I/O when the device is out of sync. This may be useful for counter applications when the network is powered down often.

SDI-12 Configuration

SDI-12 Input—Manually edit the result register parameters.

Address—Enter the device ID for the SDI-12 device.

Command—Select the command type that applies to your SDI-12 sensor.

Registers—Enable each used registers and set the Register Options that apply to your SDI-12 sensor.

Signal Conditioning

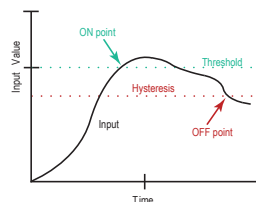
Delta

The delta parameter defines the change required between sample points of an analog input before the analog input reports a new value. To turn off this option, set the Delta value to 0. To use the delta function, the push registers must be defined.

Hysteresis and Threshold (Analog)

A system's **threshold** and **hysteresis** work together to establish the ON and OFF points of an analog input. The threshold defines a trigger point or reporting threshold (ON point) for a sensor input. When the input value is higher than the threshold, the input is ON. Hysteresis defines how far below the threshold the analog input is required to be before the input is considered OFF. A typical **hysteresis** value is 10% to 20% of the unit's range.

In the example shown, the input is considered on at 15 mA. To consider the input off at 13 mA, set the hysteresis to 2 mA. The input will be considered off when the value is 2 mA less than the threshold.



Switch Power Settings

Configure the following parameters for the **switch power** settings (inputs).

Enabled

Associates I/O switch power functions to a specific input. Do not use these parameters to configure continuous, or device-level, switch power. Select one of the available switch power (SP) checkboxes to link that switch power to the input you're currently configuring.

Voltage and Warm-Up

Select the desired voltage and warm-up time. The voltage setting establishes the voltage of the switch power. The warmup time is the length of time the switch power must be on before the device can sample the input.

Extended Input Read Settings

Use the Extended Input Read parameter to link multiple input sampling times together when all devices are powered by the same Switch Power. For battery-powered devices, this uses less energy and prolongs battery life. Define the Sample Rate, Warm-up time, and Voltage parameters for the first input. These parameters for follow-on linked inputs are ignored. Click on the **Extended Input Read** arrow to select the additional inputs to read when the switch power is active.

For example, set the Sample Rate, Warm-Up time, and Voltage for Universal Input 1. Select SP1 to power Universal Input 1. By selecting Extended Input Read for 2 and 3, the device will also read inputs 2 and 3 when it reads input 1.

Temperature Settings

Enable full scale (Temperature)

Turning Fullscale OFF sets the register range of 0x8000 (–32767) through 0x7FFF (+32768) to represent the range of input values. With Fullscale turned OFF, a register value of 1450 represents 72.5 degrees (register values = temperature × 20). With Fullscale turned ON, users can specify the register minimum and maximum range of values. These min/max values are represented in the register as 0 (min) and 65535 (max).

Temperature resolution

Select high to store temperatures values in the registers as the measured temperature × 20. Set to low to store temperature values in tenths of a degree (measured temp × 10).

For example, if the measured temperature is 20.5 degrees, turning temperature scaling to high stores the temperature value as 410 while use low resolution stores the temperature as 205.

Units

Select either Celsius or Fahrenheit for your temperature readings.

Outputs

The following parameters are used to configure the outputs, including the switch power outputs.

The output parameters vary, depending on the enabled and available output types. To begin configuring your outputs, **Enable** the output.

Discrete Output

Default Output Value

Select the **default output value**. When the selected default condition occurs and Hold Last State parameter is set to OFF, this output is set to the selected default output value (e.g. out of sync, communication timeout, start up).

Use I/O Pulse Pattern

To use a programmed pulse pattern, select **Use I/O pulse pattern**, then select the appropriate pattern from the drop-down list. Define the patterns in the Device Parameters section of this screen.

Hold Last State

Retains its last value during the selected default condition (out of sync, communication timeout, start up).

Switch Power Output

When linking a **switch power** output to a specific input, select the **Enable** checkbox and set the **Enable default state** to OFF. Use the settings for the specific input to link the switch power output and set the voltage and warm-up time.

For continuous switch power, set the voltage on this screen and set the default state to ON. Verify the default "start-up" conditions are set in the device parameters screens.

Enable default state

When enabled, this switch power output remains on during the selected default condition (e.g. out of sync, communication timeout, start up). When disabled, the switch power cycles off during the selected default condition.

Use I/O Pulse Pattern

To use a programmed pulse pattern, select **Use I/O pulse pattern**, then select the appropriate pattern from the drop-down list. Define the patterns in the Device Parameters section of this screen.

Hold Last Voltage

When set, the switch power output retains its last value during the selected default condition (e.g. out of sync, communication timeout, start up).

Voltage

To set a voltage for the switch power output, select a value. When configured for continuous voltage output, this switch power output no longer cycles on, warms up the sensors, then cycles back down. Because the output voltage remains constant, continuous voltage is typically used with solar power installations.

User Defined Devices Screen

If your device does not appear in the Device type drop-down list, it is a user-defined device. Use the **User Defined Devices** screen to define the device or to upload a configuration file specific to your device. This is particularly useful for special, non-standard MultiHop devices.

Reprogram Device Screen

Use the **Reprogram Device** screen to update the firmware and EEPROM files.

Before updating the program files, verify you are using RS-485 to communicate with the MultiHop radio. The **Reprogram Device** screen only works when the M-Hx models are in RS-485 communication mode. The communication model is selectable using jumpers for the M-H model, but is hardwired to RS-485 for the M-Hx models.

Auto Boot

Lets the configuration tool manage the boot loading process. This is the recommended setting. See ["Update Firmware Using Auto Boot" on page 25.](#)

Manual Boot

Requires the user to manually perform the beginning of the boot load process. With the device turned off, start the boot load process by clicking **Write file to device** with all the appropriate parameters defined. Turn on the power to the MultiHop device within 15 seconds, and the device should go into boot mode. Manual mode was created to deal with varying startup timing with USB ports. See ["Update Firmware Using Manual Boot" on page 25.](#)

Check part number when programming

Select **Check part number when programming** to have the Configuration Tool automatically verify the firmware or EEPROM file features match the features of your radio. For example, selecting this option allows the Configuration Tool to verify you aren't trying to write the firmware or EEPROM for an I/O radio to a data radio that does not have I/O.

Command sequence character

AT command delimiter. Most MultiHop radios require the # character. The % character is used for the DXM Controller I/O board.

Register View Screen

Use the **Register View** screen to read and write register contents when operating in Modbus mode.

The values stored in the selected registers displays on the screen.

Settings Screen

Use the **Settings** screen to send AT commands to the MultiHop radio when using the RS-485 to USB adapter cable (not over a TCP connection) or to set the ISM radio ID of the MultiHop radio within a DXM Controller.

AT Commands are control commands that access system parameters. This feature is **only for advanced users**. Do not use these commands unless you understand the intended operation. See ["Send AT Commands" on page 23](#).

To assign a **New ISM radio ID** to a MultiHop radio within a DXM Controller, enter the new radio id and click **Set ID**.

Chapter 3 Configuration Instructions

Conduct a MultiHop Site Survey

To conduct a MultiHop site survey using the configuration software, use the **Network > Network and Device Overview** screen.

To conduct a signal strength analysis, or [site survey](#), follow these instructions.

1. On the MultiHop Configuration Software, go to the **Network > Network and Device Overview** screen.
2. Select the client radio's [device ID](#) from the drop-down list.
3. Initiate a site survey analysis.
 - To analyze the entire network, select **Site survey**, then click **Read**.
 - To analyze an individual radio, highlight the line of the device then right-click and select site survey.

Depending on the size of your network, analyzing the signal strength and displaying the network structure may take a few minutes (about 15 seconds per radio.)

4. Read ["Interpreting the MultiHop Site Survey Results" on page 18](#) for detailed instructions about interpreting your results.

Signal Strength—The number of packets successfully transmitted, during a [site survey](#), at the various signal strengths are represented by the "green," "yellow," and "red" delineations.

Green—a strong radio signal

Yellow—a good radio signal

Red—a marginal radio signal

Misses—represents the number of packets not received on their first transmission

Interpreting the MultiHop Site Survey Results

Site survey mode works by having two radios (one child and one parent) repeatedly exchange data packets. For every round-trip exchange of data, the child data radio keeps track of the weaker of the two paths. Both units report the statistics as a percentage on their LCD display.

The reports consists of sorting the data into one of four categories: Green, Yellow, Red, or Missed Packets.

- Green indicates strong signal,
- Yellow is less strong but still robust,
- Red means the packet was received but has a margin of less than 15 dB, and
- A missed packet means the data did not arrive or contained a checksum error. (During normal operation, missed packets are re-tried until they are received without errors. During a site survey, missed packets are not re-tried.)

For applications with only a few hops, the system can tolerate up to 40% missed packets without serious degradation, but situations with more missed packets should be reviewed for proper antenna selection and placement, cabling, and transmit power levels. If your application includes many hops, modify the installation and antenna placement to reduce the missed packet count.

Any radio can initiate a site survey. Other radios on the same network ID remain synchronized to the network, but are blocked from sending data while the site survey is running. In installations with multiple child radios, the site survey analyzes the signal strength between the selected child and its parent radio only. Disable site survey on one radio before initiating it from another.

Radios in site survey mode have a solid green LED for the duration of the site survey and the LCD display scrolls the results. Because the statistics represent the lesser of the round-trip results, one person can ascertain the link quality from either device.

Improving Your Site Survey Results within a MultiHop Network

If a repeater radio is available in the network but is not being used, enable the forced routing function on the radio with a weak signal to force it to use a nearby radio with a stronger signal strength. See [Forced Routing with MultiHop Radios](#) for more information.

If you cannot use forced routing or add a repeater radio to the network, use an 8 dBi omni-direction antenna or a 10 dBd directional antenna.

We also recommend raising the radio units to a higher elevation, either by physically moving the devices or installing the antenna(s) remotely at a higher position. Additional antenna cables are available from Banner Engineering if needed.

The absence of signals may also be because of the distance between the client (main) and server (remote) radios. If this is the case, please contact Banner Engineering for further assistance.

Configure a User-Defined Device

To configure a device that does not appear in the **Device type** drop-down list, use the **Configuration > User Defined Devices** screen.

The screenshot displays the 'User Defined Devices' configuration window. On the left sidebar, 'Configuration' is selected. The main window has a 'Configure Device' header and a 'User Defined Devices' tab. Below this, there's a 'Custom Devices' section with a 'File' area containing 'Load File', 'Clear File', and 'Save file' buttons. A 'Devices' section has 'Add Device' and 'Remove Last Device' buttons. A dropdown menu shows 'TestDevice' with a count of 0 and an 'Add Part Number' button. Below this are three tables: 'Switch Power' (SP1, SP2, SP3, SP4 with 'Present' checkboxes), 'Inputs' (IO Point 1-14 with Input Type and Temperature dropdowns), and 'Outputs' (IO Point 1-15 with Output Type dropdowns).

1. Name your new device template. In the example shown, we have named our template TestDevice.
2. Enter the part number of your device and click **Add Part Number**.
3. If your device includes switch power outputs, select the active **switch power** outputs.
4. Define each input and output type.
5. Click **Save File** to save the file to your hard drive.
6. Click **Load File** to load the file to your device.

Inputs and Outputs

Configure the Switch Power Output

To operate the [switch power](#) (SP) terminals as switch power that cycles on for a specific length of time then cycles off to conserve power, enable the desired SP terminal and set the voltage and warm-up time for the specific input associated with the switch power. Follow these steps to associate SP3 to Analog IN 1.

The screenshot shows the configuration interface for Analog Input 1 and Switch Power Output 3. On the left, under 'Analog Input 1', the 'Enabled' checkbox is checked. Below it, 'Analog Configuration' shows 'Enable push on change of state' and 'Enable full scale' checked, with 'Maximum value' set to 20000 and 'Minimum value' set to 0. 'Signal Conditioning' has 'Threshold', 'Hysteresis', and 'Delta' all set to 0. 'Switch Power Settings' shows 'SP3' checked, 'Warm up' set to 00:00:00.12, and 'Voltage' set to 10V. 'Extended Input Read' is expanded. 'Sample Out of Sync' has 'Continue sampling when out of sync' checked. On the right, under 'Switch Power Output 3', the 'Enabled' checkbox is checked. 'Output Configuration' has 'Output operates when radio is out of sync' checked and 'Use flash pattern' unchecked. 'Switch Power Options' has 'Hold last voltage' unchecked and 'On signal loss, output is active' checked. 'Voltage' is set to 10V.

1. Go to the Analog IN 1 section of the **Configuration > Configure Device** screen and select the **Enabled** checkbox.
2. Select the **SP3** checkbox to enable switch power 3 to power analog IN 1.
3. Set the voltage and warm-up time according to the sensor's needs.
4. Use the **Switch Power Output 3** section to enable the switch power.
5. Set the appropriate Output Configuration and Switch Power Option.

Configure Device-Level Switch Power

Use device-level [switch power](#) to configure voltage output through the switch power terminals. If switch power is defined for use in any input point, switch power cannot be defined for device-level use.

Device-level switch power is configured using the Outputs pane of the I/O parameters.

1. In the MultiHop Configuration Tool, go to the **Configuration > Configure Device** screen.
2. Enter the **Device ID** and **Device type**.
3. Click **GET all parameters and I/O** to retrieve the configuration parameters from the device.
4. On the Output side of the screen, click the arrow next to the **Switch Power Output** to display the parameters.
5. Enable the switch power.
6. Define the **Voltage** and select a flash pattern if applicable.
7. Click **SEND** to send the settings to the radio.
8. After sending any changes to the radio, cycle the power to activate the changes.

The screenshot shows the configuration for Switch Power Output 3. The 'Enabled' checkbox is checked. 'Output Configuration' has 'Use flash pattern' unchecked and a value of 1. 'Switch Power Options' has 'Hold last voltage' unchecked and 'Enable default state' checked. 'Voltage' is set to 12.

Configure Switch Power for Continuous Voltage

To configure the Switch Power output to supply continuous voltage to a sensor using the configuration software, follow these steps.

On your computer, launch the MultiHop Configuration Software. Select the appropriate COM connection to the computer.

This example sets up switch power 1 to supply continuous power. Note that only one switch power output can be configured to supply continuous power, and no input can be defined to use switch power.

1. On the **Configuration > Configure Device** screen, enter the ID of the MultiHop radio that will be assigned a continuous power output and click **GET all parameters and I/O**.

The configuration of that MultiHop radio is read from the radio and displayed on the screen.

2. Select the **Switch Power Output** and enable it.
3. Select **Enable default state**.
4. Use the drop-down list and select a value for the **Voltage**.
5. Click **SEND**.
6. Under the **Default Conditions and Power Save** settings, select **Start Up**.

This configures the enabled and configured Switch Power terminal to output continuous voltage as soon as the radio powers up.

7. Click **SEND system parameters** or **SEND all parameters and IO** to send the new settings to the wireless network device.
8. Cycle power to the radio.

This continuous **switch power** configuration will activate after the power is cycled.

Configure the SDI-12 Inputs

To configure the SDI-12 inputs for use with an Acclima or Decagon 5TE SDI-12 sensor, follow these steps.

1. Select the **Enable** checkbox to enable the SDI-12 input.
2. Select an **Sample Interval** (sample rate), in hours:minutes:seconds.
3. Select the **Address** (device ID) for the SDI-12 device. The sample screenshot shows the configuration for SDI-12 device 1.
4. Under the **Switch Power Settings**, select either 1 or 2 to have SP1 or SP2 power your SDI-12 device.
5. Enter the appropriate **Voltage** needed to power your sensor.
6. Select a **Warm-Up** time, in hours:minutes:seconds. This is the length of time your sensor requires power before an accurate reading occurs. Refer to your SDI-12 documentation for this value.
7. If you want your SDI-12 device to continue sampling after the MultiHop radio has lost its radio connection to its parent radio, select the **Enable Input Out-of-Sync** checkbox.

8. Enable each required register and select the appropriate register option from the drop-down list.
9. Click on **Send** to send this configuration to the selected MultiHop device.

Push Registers

Configure the Push Registers

Up to four push registers may be defined for each MultiHop radio. For most applications, input registers are mapped to the push registers, but other registers may be configured as push registers, such as the battery read voltage. To configure a register as a push register:

1. On the MultiHop Configuration Tool, go to the **Configuration > Configure Device** screen.
2. Select the desired MultiHop Radio ID number at the top of the screen.
3. Click the arrow next to **Push Registers** to display the push registers parameters.
4. Use the **Registers** drop-down list to select the input for that radio that will be a push register.
5. Enter a **Push Interval** or **client Sweep Interval** value.
6. Click **Send** to write the settings to the radio.

The status bar at the bottom of the screen indicates when the process is complete by listing the Communication Status as Ready, with a green light. Errors display as a red light and a brief explanation of the error status. A yellow status light and a description of Getting Data indicate the network is retrieving the requested data from the selected MultiHop radio.

Push Register Index	
1	0
2	0
3	0
4	0

Push Interval

Enter a value in seconds to define how often the push register values are transmitted back to local registers on the radio client.

Client Sweep Interval (Radio client only)

Enter a value in seconds to define how often the client should expect each repeater and server to push register values back to the cached copies in local storage on the radio client. The client Sweep Interval is only set on the radio client and should always be greater than the push interval defined for the other radios. When the radio client does not receive a push register value within the client Sweep Interval, the status register for that MultiHop radio is set to indicate there has been no communication from that radio.

Reading Push Registers

To read the push register data from the client device, send a Modbus read command to **the remote server ID for registers 47909-47912** (push registers 1-4). The client device intercepts the Modbus command to the remote server (because of the register addresses) and substitutes the local cached register values.

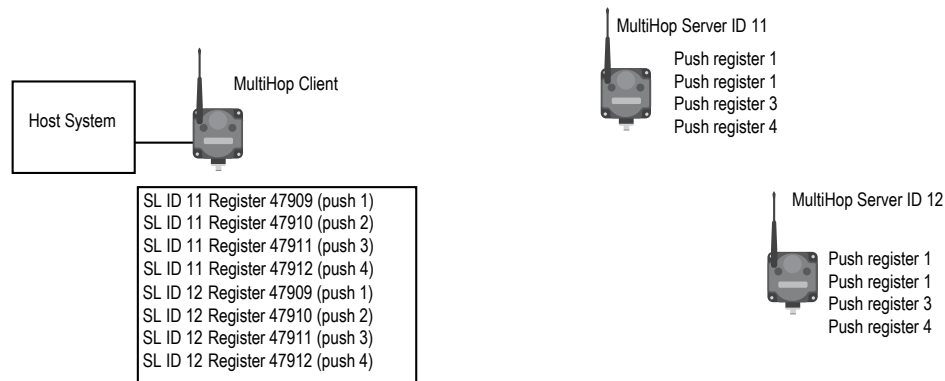
Read the status register for a MultiHop device the same as reading the push registers: the host sends a Modbus read command to the remote server ID and register 47904. To use the status register for each MultiHop device, the client sweep interval must be defined using the MultiHop Configuration tool.

A status register value of zero indicates no device; a value of one (1) indicates the device has reported.

Example Push Registers

When the Modbus client/host system needs I/O data, it queries the MultiHop radio client. The radio client “intercepts” the request and sends the data to the Modbus client/host system without having to wirelessly request the data from the MultiHop radio server. This allows a host system to interrogate only the radio client for I/O information and avoid the latency of the wireless link.

For example, when the host system requests the contents of a particular MultiHop radio server's registers 47909 through 47912, the radio client “intercepts” the request and pulls the appropriate values from its stored registers.



Using Change of State and Push Registers

Push registers are typically defined to send back register data at specific time intervals. Change of state data can also be defined to send to the radio client. As with any push register data, the data is not guaranteed to be delivered to the radio client because the message is not acknowledged from end to end.

The factory default for message retries is set to eight. With really poor links, the data may not be received. We recommend also having a periodic reporting time defined when using [change of state](#).

Read Registers

Use the **Register View** screen to read registers.

1. On the **Register View** screen, select the MultiHop Radio ID of the desired radio.
2. Under **Data Format**, select how to display the registers values, either decimal or hexadecimal.
3. Under **Read Registers**, select the starting register and the number of registers from that starting point to read. In the sample screen shown, register 1 and the 5 following registers will be read and displayed on the screen.
4. Select **Enable Polling** and enter a polling frequency in seconds.
5. Click **Read Registers**.

The status bar at the bottom of the screen indicates when the process is complete by displaying a communication status of Ready and a green light.

Write to Registers

Use the **Register View** screen to write to selected registers.

1. On the **Register View** screen, select the MultiHop Radio ID of the desired radio.
2. Under **Write Registers**, select the starting register and the number of registers from that starting point to read.
3. Enter the values, in hex, to be written to each register.
4. Click **Write Registers** to send these changes to the radio.

The status bar at the bottom of the screen indicates when the process is complete by displaying a communication status of Ready and a green light.

Send AT Commands

To send AT commands to the MultiHop radio, use the **Settings** screen.

1. Enter the command in the first field, just to the right of the **AT**.
2. Enter the value in the second field, near the **Send AT command** button.
3. Click **Send AT command**. The status bar at the bottom of the screen indicates **Ready** when the process is complete. When the AT command was successfully sent to the radio, the feedback field displays **OK**.

Using the MultiHop Configuration Software with the MultiHop Ethernet Radio

The MultiHop Configuration Software can be used with the Ethernet Data Radios to examine the network topology, conduct a [site survey](#), or adjust parameter settings.

The MultiHop Configuration Software must be set up to communicate through the Ethernet connection by using the IP address of the client MultiHop Ethernet Data Radio. To use the MultiHop Configuration Software with the MultiHop Ethernet Data Radio, the computer must be plugged into the same wired network as the client radio.

The default IP address of the Ethernet radios is 192.168.17.17. After connecting, set the software's addressing mode to use the [device ID](#) of the Ethernet radio.

1. Launch the MultiHop Configuration Software.
2. Go to the **Device > Connection Settings** menu.
3. Select **TCP DXM** and enter the IP address of the client Ethernet radio. Communications timeout and communication retry parameters can also be adjusted on this page.
4. Click **Connect**.
5. Go to the **Network > Network and Device Overview** screen.
6. Enter the client Device address into the **Device address** field. From the LCD menu, the device address is shown as (DADR).
7. Click **Read** to view the wireless network.

To conduct a signal strength analysis, or [site survey](#), select **Site survey**, then click **Read**. Depending on the size of your network, analyzing the signal strength and displaying the network structure may take a few minutes (about 15 seconds per radio.)

To run site survey on a individual radio, highlight the line of the device then right click.

Depending upon the radio link quality, the retry count and timeout parameters may need to be adjusted. Those parameters are found under the **Device > Connection Settings** menu.

Name	Role	Modbus Address	Device Address	Parent Address	Signal Strength	Green	Yellow	Red	Misses	Serial Number	Model Number	Build Date	RF FW PN	RF EE Ver	RF EE PN	RF EE Ver	LCD FW PN	LCD EE Ver	LCD EE PN	LCD EE Ver
IPADDR192168017017	Master	11	64133	64133	0	0	0	0	0	129669	166172	1517	162685	3.5	162687	1.1	136499	3.4	145387	1.0
IPADDR192168017016	Slave	12	14496	64133	0	0	0	0	0	276630	166172	1426	162685	3.1	162687	1.1	136499	3.1	145387	1.0

Chapter 4 Product Support

View Device Factory Information

Use the **Network > Network and Device Overview** screen to view the device's factory information.

Network

Configuration

Reprogram

Register View

Settings

Network and Device Overview

Network Query

Master address1Device addressReadSite Survey

Devices: 24Repeater: 1Slaves: 22Unreachable: 2

Save to File

Name	Role	Modbus Address	Device Address	Parent Address	Signal Strength	Green	Yellow	Red	Misses	Serial Number	Model Number	Build Date	RF FW Ver	RF EE Ver	RF EE Ver	LCD FW Ver	LCD EE Ver	LCD EE Ver		
Master 900MHz HES	Master	1	23846	23846	0	0	0	0	0	154918	196215	001544	175068	3.6C	175070	1.0				
DATA RADIO DEVICE	Slave	35	34520	23846	50	0	0	0	50	100056	000000	000000	165062	3.0E	159481	0.2A				
DATA RADIO DEVICE	Slave	17	24200	23846	0	0	0	0	0	155272	151687	001544	169893	3.4	157721	1.1				
Multi Data Radio	Slave	14	64179	23846	0	0	0	0	0	195251	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	45	63129	23846	0	0	0	0	0	259737	151687	001415	169893	2.6	157721	1.1				
DATA RADIO DEVICE	Slave	19	24203	23846	0	0	0	0	0	155275	151687	001544	169893	3.4	157721	1.1				
DATA RADIO DEVICE	Slave	90	4775	23846	0	0	0	0	0	135847	183420	001523	169893	2.6	157721	1.1				
Multi Data Radio	Slave	15	64180	23846	0	0	0	0	0	195252	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	37	56005	23846	0	0	0	0	0	842437	190055	1541	169345	3.1	169449	0.1C				
Multi Data Radio	Slave	16	64184	23846	0	0	0	0	0	195256	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	20	24196	23846	0	0	0	0	0	155266	151687	001544	169893	3.4	157721	1.1				
DATA RADIO DEVICE	Slave	36	56006	23846	0	0	0	0	0	842438	190055	1541	169345	3.1	169449	0.1C				
MH MGate SID 13	Slave	13	64176	23846	0	0	0	0	0	195248	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	18	24202	23846	0	0	0	0	0	155274	151687	001544	169893	3.4	157721	1.1				
DATA RADIO DEVICE	Slave	27	9619	23846	0	0	0	0	0	271963	151687	001425	169893	2.6	157721	1.1				
Multi Radio H12	Repeater	91	50281	23846	78	0	0	0	22	123817	151685	1512	148691	2.2	151698	1.3	136499	3.2	148880	1.0
DATA RADIO DEVICE	Slave	84	4794	50281	0	0	0	0	0	135666	183420	001523	169893	2.6	157721	1.1				
DATA RADIO DEVICE	Slave	32	9621	50281	0	0	0	0	0	271965	151687	001425	169893	2.6	157721	1.1				
MH MGate SID 12	Slave	12	64185	50281	0	0	0	0	0	195257	157598	001233	157719	2.2	157722	1.0				
Multi Data Radio	Slave	78	29005	50281	0	0	0	0	0											
DATA RADIO DEVICE	Slave	31	65199	50281	0	0	0	0	0	261906	151687	001417	169893	2.6	157721	1.1				
DATA RADIO DEVICE	Slave	82	4744	50281	0	0	0	0	0	135816	183420	001523	169893	2.6	157721	1.1				
MH MGate SID 11	Slave	11	64181	50281	0	0	0	0	0	195253	157598	001233	157719	2.2	157722	1.0				
DATA RADIO DEVICE	Slave	83	4743	50281	0	0	0	0	0	135815	183420	001523	169893	2.6	157721	1.1				
Unreachable devices addresses																				
Device Address: 477529001Reprocess																				

1. To view the factory information for all models within your network, such as model and serial numbers, select the client radio's ID and click **Read**.
2. To view the factory information for a specific device, click on the device name and select from the context-sensitive menu.

Update Firmware Using Auto Boot

Update firmware or EEPROM files from the **Program > Reprogram Device** screen.

To write a new version of the firmware or EEPROM files to the device using **Auto Boot** mode, follow these instructions. Do not attempt any other work on the computer while updating program files to a MultiHop radio as this may corrupt the programming process.

1. Click **Write File to Device**.
2. Select the new file from your hard drive.
3. A dialogue box verifies that you want to overwrite the configuration file already on the data radio. Click **OK** to continue or **Cancel** to cancel the write process.
4. The Communication Status display at the bottom of the screen indicates when the software finishes writing the file to the radio. When the status is **Ready**, the process is complete. If there are errors during the programming process, use manual mode to update the files.

To save your firmware or EEPROM files to your hard drive using **Auto Boot** mode, select a location and file name on your computer.

Update Firmware Using Manual Boot

Update firmware or EEPROM files from the **Program > Reprogram Device** screen.

To write a new version of the firmware or EEPROM files to the selected device using **Manual Boot** mode, follow these instructions. Do not attempt any other work on the computer while updating program files to a MultiHop radio as this may corrupt the programming process.

1. Cycle power to the radio by disconnecting then reconnecting the power to the radio.
The firmware version will be read when the radio reboots.

2. After the radio reboots, browse to the firmware file location on your hard drive and select the appropriate files.
3. Click **Write File to Device** for the firmware, then click **OK** on the pop-up box to start the process.
4. Immediately cycle power to the MultiHop radio.
5. Complete the same steps for the EEPROM if necessary.

To save your firmware or EEPROM files to your hard drive using **Manual Boot** mode:

1. Cycle power to the radio by disconnecting then reconnecting the power to the radio. The firmware version will be read when the radio reboots.
2. After the radio reboots, browse to the firmware file location on your hard drive. Select the appropriate files.
3. Click **Write File to Device** for the firmware, then click **OK** on the pop-up box to start the process.
4. Immediately cycle power to the MultiHop radio.
5. Complete the same steps for the EEPROM if necessary.

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Glossary

change of state

A change of state (COS) is when an input signal switches from Run-to-Stop or Stop-to-Run state. Dual-channel input signals, have two possible configurable COS settings describe the signal disparity limits that can exist between channels before a fault condition is registered: simultaneous and concurrent. If simultaneity is a requirement or a concern for the application, users must ensure they made the correct selection during configuration.

default output value

Default output values are specific values written to output registers. For discrete outputs, the value is a 1 (on) or 0 (off) value. For analog outputs, the value can be any valid register value. When a default condition occurs, these default output values are written to the output register.

device ID

The device ID is a unique identifier for each device on a network. For the DX80 wireless star networks, Gateways are identified as device 0 and Nodes are assigned an ID from 01 to 47 using the rotary dials. For MultiHop networks, device IDs 1-10 are reserved for servers directly connected to the host. Use device IDs 11-60 for remote Modbus servers (devices serially connected to a data radio) for a maximum of 50 devices.

event counter

The event counter counts the total number of times an input signal changes to the high/ON/1 state. The counter increments on the falling edge of an input signal when the signal level crosses the threshold. Event counters can be used to measure the total operational cycles of a spinning shaft or the total number of items traveling down a conveyor.

flash pattern

Flash patterns are established by selecting timeslots to turn the output on or off. Although the flash pattern was originally designed to turn on and off an indicator light, a flash pattern can be set for any discrete output or switch power output.

frequency counter

The frequency counter calculates the frequency of the input signal, in Hz. Frequency counters can be used to measure flow rates, such as measuring the flow rate of items on a conveyor or the speed at which a windmill spins.

hysteresis

Hysteresis defines how far below the active threshold (ON point) an analog input is required to be before the input is considered OFF. A typical hysteresis value is 10% to 20% of the unit's range.

median filter

When the median filter is turned on, three samples are taken for each analog sensor reading. The high and low values are discarded and the middle value is used as the analog value. Set to zero (0) to turn off the median filter. Set to one (1) to turn on the median filter.

out of sync radio link loss

The Sure Cross wireless devices use a deterministic link time-out method to address radio frequency (RF) link interruption or failure. When a radio link fails, all pertinent wired outputs are sent to the selected default value/state until the link is recovered, ensuring that disruptions in the communications link result in predictable system behavior. Following a time-out, all outputs linked to the Node in question are set to 0, 1, or hold the last stable state, depending on the value selected.

report interval or report rate

The report rate defines how often the Node communicates the I/O status to the Gateway. For battery-powered applications, setting the report rate to a slower rate extends the battery life.

sample interval or sample rate

The sample interval, or rate, defines how often the Sure Cross device samples the input. For battery-powered applications, setting a slower rate extends the battery life.

site survey

Conducting a site survey, also known as a radio signal strength indication (RSSI), analyzes the radio communications link between the client radio and any server radio within the network by analyzing the radio signal strength of received data packets and reporting the number of missed packets that required a retry.

switch power

Efficient power management technology enables some FlexPower devices to include an internal power output supply, called switch power (SP), that briefly steps up to power sensors. The warmup time denotes how long the sensor must be powered before a reliable reading can be taken. After the warmup time has passed, the input reads the sensor, then the switched power shuts off to prolong battery life.

threshold and hysteresis

Threshold and hysteresis work together to establish the ON and OFF points of an analog input. The threshold defines a trigger point or reporting threshold (ON point) for a sensor input. When the input value is higher than the threshold, the input is ON. Hysteresis defines how far below the threshold the analog input is required to be before the input is considered OFF. A typical hysteresis value is 10% to 20% of the unit's range. Setting threshold and hysteresis points prevents inputs from oscillating between 'on' and 'off' when the input remains close to the threshold point. In the example shown, the input is considered on at 15 mA. To consider the input off at 13 mA, set the hysteresis to 2 mA. The input will be considered off when the value is 2 mA less than the threshold.

timeout interval

The timeout interval is the total elapsed time before the system flags an error condition. This is a calculated value from polling interval (sec) × maximum misses.

